
Lecture Notes for Chapter 3:

Growth of Functions

Chapter 3 overview

- A way to describe behavior of functions *in the limit*. We're studying **asymptotic** efficiency.
- Describe *growth* of functions.
- Focus on what's important by abstracting away low-order terms and constant factors.
- How we indicate running times of algorithms.
- A way to compare "sizes" of functions:

$$O \approx \leq$$

$$\Omega \approx \geq$$

$$\Theta \approx =$$

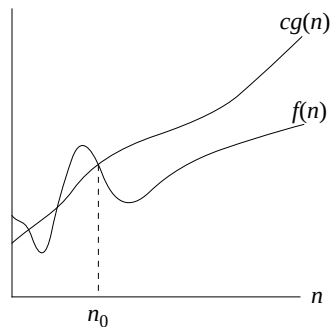
$$o \approx <$$

$$\omega \approx >$$

Asymptotic notation

***O*-notation**

$O(g(n)) = \{f(n) : \text{there exist positive constants } c \text{ and } n_0 \text{ such that}$
 $0 \leq f(n) \leq cg(n) \text{ for all } n \geq n_0\}$.



$g(n)$ is an **asymptotic upper bound** for $f(n)$.

If $f(n) \in O(g(n))$, we write $f(n) = O(g(n))$ (will precisely explain this soon).

Example

$2n^2 = O(n^3)$, with $c = 1$ and $n_0 = 2$.

Examples of functions in $O(n^2)$:

$$n^2$$

$$n^2 + n$$

$$n^2 + 1000n$$

$$1000n^2 + 1000n$$

Also,

$$n$$

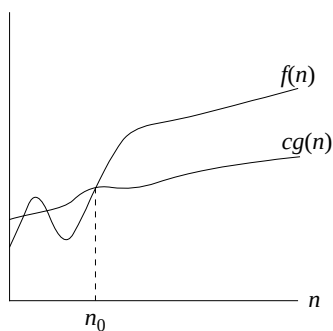
$$n/1000$$

$$n^{1.99999}$$

$$n^2 / \lg \lg \lg n$$

 Ω -notation

$\Omega(g(n)) = \{f(n) : \text{there exist positive constants } c \text{ and } n_0 \text{ such that}$
 $0 \leq cg(n) \leq f(n) \text{ for all } n \geq n_0\}$.



$g(n)$ is an **asymptotic lower bound** for $f(n)$.

Example

$\sqrt{n} = \Omega(\lg n)$, with $c = 1$ and $n_0 = 16$.

Examples of functions in $\Omega(n^2)$:

$$n^2$$

$$n^2 + n$$

$$n^2 - n$$

$$1000n^2 + 1000n$$

$$1000n^2 - 1000n$$

Also,

$$n^3$$

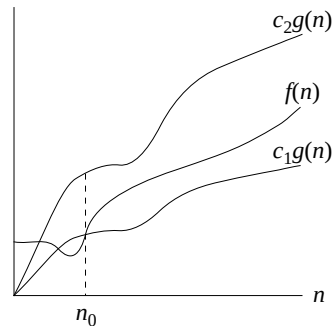
$$n^{2.00001}$$

$$n^2 \lg \lg \lg n$$

$$2^{2^n}$$

Θ -notation

$\Theta(g(n)) = \{f(n) : \text{there exist positive constants } c_1, c_2, \text{ and } n_0 \text{ such that}$
 $0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n) \text{ for all } n \geq n_0\}.$



$g(n)$ is an **asymptotically tight bound** for $f(n)$.

Example

$n^2/2 - 2n = \Theta(n^2)$, with $c_1 = 1/4$, $c_2 = 1/2$, and $n_0 = 8$.

Theorem

$f(n) = \Theta(g(n))$ if and only if $f = O(g(n))$ and $f = \Omega(g(n))$.

Leading constants and low-order terms don't matter.

Asymptotic notation in equations**When on right-hand side**

$O(n^2)$ stands for some anonymous function in the set $O(n^2)$.

$2n^2 + 3n + 1 = 2n^2 + \Theta(n)$ means $2n^2 + 3n + 1 = 2n^2 + f(n)$ for some $f(n) \in \Theta(n)$. In particular, $f(n) = 3n + 1$.

By the way, we interpret # of anonymous functions as = # of times the asymptotic notation appears:

$$\sum_{i=1}^n O(i) \quad \text{OK: 1 anonymous function}$$

$$O(1) + O(2) + \cdots + O(n) \quad \text{not OK: } n \text{ hidden constants}$$

$\Rightarrow$ no clean interpretation

When on left-hand side

No matter how the anonymous functions are chosen on the left-hand side, there is a way to choose the anonymous functions on the right-hand side to make the equation valid.

Interpret $2n^2 + \Theta(n) = \Theta(n^2)$ as meaning for all functions $f(n) \in \Theta(n)$, there exists a function $g(n) \in \Theta(n^2)$ such that $2n^2 + f(n) = g(n)$.

Can chain together:

$$\begin{aligned} 2n^2 + 3n + 1 &= 2n^2 + \Theta(n) \\ &= \Theta(n^2) . \end{aligned}$$

Interpretation:

- First equation: There exists $f(n) \in \Theta(n)$ such that $2n^2 + 3n + 1 = 2n^2 + f(n)$.
- Second equation: For all $g(n) \in \Theta(n)$ (such as the $f(n)$ used to make the first equation hold), there exists $h(n) \in \Theta(n^2)$ such that $2n^2 + g(n) = h(n)$.

***o*-notation**

$o(g(n)) = \{f(n) : \text{for all constants } c > 0, \text{ there exists a constant } n_0 > 0 \text{ such that } 0 \leq f(n) < cg(n) \text{ for all } n \geq n_0\}$.

Another view, probably easier to use: $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$.

$$\begin{aligned} n^{1.9999} &= o(n^2) \\ n^2 / \lg n &= o(n^2) \\ n^2 &\neq o(n^2) \text{ (just like } 2 \not\prec 2) \\ n^2 / 1000 &\neq o(n^2) \end{aligned}$$

***ω*-notation**

$\omega(g(n)) = \{f(n) : \text{for all constants } c > 0, \text{ there exists a constant } n_0 > 0 \text{ such that } 0 \leq cg(n) < f(n) \text{ for all } n \geq n_0\}$.

Another view, again, probably easier to use: $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$.

$$\begin{aligned} n^{2.0001} &= \omega(n^2) \\ n^2 \lg n &= \omega(n^2) \\ n^2 &\neq \omega(n^2) \end{aligned}$$

Comparisons of functions

Relational properties:

Transitivity:

$$f(n) = \Theta(g(n)) \text{ and } g(n) = \Theta(h(n)) \Rightarrow f(n) = \Theta(h(n)).$$

Same for O , Ω , o , and ω .

Reflexivity:

$$f(n) = \Theta(f(n)).$$

Same for O and Ω .

Symmetry:

$$f(n) = \Theta(g(n)) \text{ if and only if } g(n) = \Theta(f(n)).$$

Transpose symmetry:

$$\begin{aligned} f(n) &= O(g(n)) \text{ if and only if } g(n) = \Omega(f(n)). \\ f(n) &= o(g(n)) \text{ if and only if } g(n) = \omega(f(n)). \end{aligned}$$

Comparisons:

- $f(n)$ is **asymptotically smaller** than $g(n)$ if $f(n) = o(g(n))$.
- $f(n)$ is **asymptotically larger** than $g(n)$ if $f(n) = \omega(g(n))$.

No trichotomy. Although intuitively, we can liken O to $\leq$, Ω to $\geq$, etc., unlike real numbers, where $a < b$, $a = b$, or $a > b$, we might not be able to compare functions.

Example: $n^{1+\sin n}$ and n , since $1 + \sin n$ oscillates between 0 and 2.

Standard notations and common functions

[You probably do not want to use lecture time going over all the definitions and properties given in Section 3.2, but it might be worth spending a few minutes of lecture time on some of the following.]

Monotonicity

- $f(n)$ is **monotonically increasing** if $m \leq n \Rightarrow f(m) \leq f(n)$.
- $f(n)$ is **monotonically decreasing** if $m \geq n \Rightarrow f(m) \geq f(n)$.
- $f(n)$ is **strictly increasing** if $m < n \Rightarrow f(m) < f(n)$.
- $f(n)$ is **strictly decreasing** if $m > n \Rightarrow f(m) > f(n)$.

Exponentials

Useful identities:

$$\begin{aligned} a^{-1} &= 1/a, \\ (a^m)^n &= a^{mn}, \\ a^m a^n &= a^{m+n}. \end{aligned}$$

Can relate rates of growth of polynomials and exponentials: for all real constants a and b such that $a > 1$,

$$\lim_{n \rightarrow \infty} \frac{n^b}{a^n} = 0,$$

which implies that $n^b = o(a^n)$.

A suprisingly useful inequality: for all real x ,

$$e^x \geq 1 + x.$$

As x gets closer to 0, e^x gets closer to $1 + x$.

Logarithms

Notations:

$$\begin{aligned}\lg n &= \log_2 n \quad (\text{binary logarithm}) , \\ \ln n &= \log_e n \quad (\text{natural logarithm}) , \\ \lg^k n &= (\lg n)^k \quad (\text{exponentiation}) , \\ \lg \lg n &= \lg(\lg n) \quad (\text{composition}) .\end{aligned}$$

Logarithm functions apply only to the next term in the formula, so that $\lg n + k$ means $(\lg n) + k$, and *not* $\lg(n + k)$.

In the expression $\log_b a$:

- If we hold b constant, then the expression is strictly increasing as a increases.
- If we hold a constant, then the expression is strictly decreasing as b increases.

Useful identities for all real $a > 0$, $b > 0$, $c > 0$, and n , and where logarithm bases are not 1:

$$\begin{aligned}a &= b^{\log_b a} , \\ \log_c(ab) &= \log_c a + \log_c b , \\ \log_b a^n &= n \log_b a , \\ \log_b a &= \frac{\log_c a}{\log_c b} , \\ \log_b(1/a) &= -\log_b a , \\ \log_b a &= \frac{1}{\log_a b} , \\ a^{\log_b c} &= c^{\log_b a} .\end{aligned}$$

Changing the base of a logarithm from one constant to another only changes the value by a constant factor, so we usually don't worry about logarithm bases in asymptotic notation. Convention is to use $\lg$ within asymptotic notation, unless the base actually matters.

Just as polynomials grow more slowly than exponentials, logarithms grow more slowly than polynomials. In $\lim_{n \rightarrow \infty} \frac{n^b}{a^n} = 0$, substitute $\lg n$ for n and 2^a for a :

$$\lim_{n \rightarrow \infty} \frac{\lg^b n}{(2^a)^{\lg n}} = \lim_{n \rightarrow \infty} \frac{\lg^b n}{n^a} = 0 ,$$

implying that $\lg^b n = o(n^a)$.

Factorials

$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$. Special case: $0! = 1$.

Can use ***Stirling's approximation***,

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + \Theta\left(\frac{1}{n}\right)\right) ,$$

to derive that $\lg(n!) = \Theta(n \lg n)$.